

Henry Brooks

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ABOUT

Computer Science MSci student at the University of Birmingham with experience building full-stack applications using Java and Spring Boot. Developed several personal and university projects involving REST APIs, CI/CD pipelines, and Docker-based deployment.

I have contributed to academic research through my Master's thesis in evolutionary game theory, focusing on computational modelling and numerical simulation of multi-agent systems. This work involves analysing emergent behaviour in complex adaptive systems.

I am currently seeking a junior/graduate software role in Denmark.

EDUCATION

University of Birmingham

Birmingham, UK

MSci Computer Science (First Class Honours, 85)

2022–2026

- Relevant modules: Object-Oriented Programming (Java), Systems Programming, Security and Networks, Software Engineering, Game Theory, Advanced Networking, Advanced Cryptography
- Master's research project in evolutionary game theory (First Class achieved, 85)
- School prize for highest grade in cohort

EXPERIENCE

University of Birmingham

Birmingham, UK

Undergraduate Teaching Assistant (Object-Oriented Programming in Java)

Aug. 2025 – Jan. 2026

- Supported teaching for an introductory Object-Oriented Programming course using Java
- Assisted students with core programming and OOP concepts
- Helped run lab sessions and provided guidance on coursework and exercises

PROJECTS

See full list of projects on henrybrooks.uk/projects

Evolutionary Game Theory Simulation System

2025 – Present

Python, NumPy, SciPy

- Developed computational models to simulate evolutionary dynamics in finite populations
- Implemented numerical simulations using NumPy and SciPy to analyse system behaviour
- Studied multi-agent interactions and probabilistic outcomes in game-theoretic systems
- Contributed to ongoing academic research in evolutionary game theory at the University of Birmingham
- Research results presented at an academic conference; abstract available: [conference abstract](#)

SAPSOC League Website

2023 – 2025

Java, Spring Boot, Angular, Docker

- Designed and developed a full-stack league management system used by a university club with 40+ active members
- Built RESTful backend services using Spring Boot and integrated Angular frontend
- Implemented CI/CD pipeline using Docker and GitHub Actions for automated deployment
- Deployed and maintained application on a self-hosted server using Nginx reverse proxy
- Reduced manual league administration time through automation and improved usability

Portfolio & League Archive Platform

2026 – Present

C#, .NET, Azure, Blazor

- Developing a backend system in C# and ASP.NET to manage and display personal projects and league data
- Deploying to Azure with CI/CD pipelines for automated deployment

AWARDS & EXTRACURRICULAR

- School Prize for best student in MSci Computer Science 2025/26
 - Awarded for achieving the highest grade in my cohort
- University Snooker Club Committee Member 2022 – Present
 - Committee member for over 3 years in a 40+ member student sports club. Contributed to the organisation and running of weekly sessions, coordinated members, and helped maintain consistent club operations and engagement.
 - Developed and maintained internal league management software used to run club league, including scheduling and tracking results. The system was actively used in real club operations.
 - Supported organisation of tournaments, and club events.
- University of Birmingham Computer Science Excellence Scholarship 2022

SKILLS

- **Languages:** Java, Python, JavaScript/TypeScript
- **Backend:** Spring Boot, REST APIs, .NET
- **Frontend:** JavaScript, HTML, CSS, Angular
- **DevOps:** Docker, GitHub Actions, Nginx
- **Cloud & Systems:** Azure (learning), Linux
- **Tools:** Git, IntelliJ, Visual Studio Code